

Financial Networks Project

Progress Report, Literature Review, and a Problem Statement
on Laplacian versus Modularity Spectra

Jash Dalal

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Abstract

Three papers on learning network structure from financial time series have been replicated from scratch over the period 5 July – 10 August 2026. This document (i) records what was built, what reproduced and what did not, in chronological order; (ii) extracts the findings that survived adversarial re-audit and are therefore usable as the foundation of new work; (iii) reviews the literature on spectral indicators of market stress, separating the *Laplacian* family (algebraic connectivity, spectral gap, Laplacian energy) from the *modularity-matrix* family (leading and trailing eigenvalues, structural eigenvalue counts, detectability thresholds); and (iv) proposes three research directions, with preliminary numerical evidence computed on the 6,315-day network sequence already built for Paper 3. The headline of that preliminary evidence: on the same data, under the same scoring protocol, the scalar modularity Q detects crises at ROC-AUC 0.538 while the spectrum of the modularity matrix it is derived from reaches 0.760, and the correlation spectrum reaches 0.822. The Fiedler value and the leading modularity eigenvalue — exact affine duals on a regular graph — are empirically *near-orthogonal* on these networks ($r = -0.17$).

How to comment. The easiest route is Overleaf’s own review mode (Review → Add comment); no macros needed. If you prefer margin notes in the source, `\pc{...}` gives a yellow margin note, `\open{...}` an orange one for unresolved questions. Section 8 collects the specific decisions I would like your view on.

Prof: replace this note with first comment check the macro renders.

Contents

1	Purpose and scope	2
2	Timeline	2
3	Part I — What was built and what it showed	3
3.1	Paper 1: learning graph Laplacians from returns	3
3.2	Paper 2: TICC	4
3.3	Paper 3: modular dynamics of financial market networks	4
4	Part II — What the three replications jointly establish	6
5	Part III — Problem statement: Laplacian versus modularity spectra	6
5.1	The two operators	6
5.2	The identity that makes the comparison non-trivial	6
5.3	Literature map	7
5.4	The gap	8
6	Part IV — Three proposed directions	8
6.1	D1 — The operator comparison: duality, and what breaks it	8
6.2	D2 — Detectability: is there community structure to detect at all?	9
6.3	D3 — The negative spectrum: anti-communities and the 2002 break	10
6.4	Recommendation	10

7	Part V — Preliminary evidence	10
8	Part VI — Decisions I would like your view on	12
	Appendix A — Environment and reproduction	12
	References	12

1 Purpose and scope

The work so far has been a *literature review conducted by replication*: rather than reading three papers and summarising them, each was rebuilt from first principles, run on real data, and audited. That was deliberate. A summary tells you what a paper claims; a replication tells you which of those claims survive contact with data, and — more useful for choosing a research problem — which of them were never actually tested by the original authors.

The output of that phase is a set of concrete, defensible gaps. This document converts them into a problem statement. The specific question on the table, raised in our last discussion, is:

Paper 1 uses the *Fiedler value* $\lambda_2(\mathbf{L})$ — the second-smallest eigenvalue of a learned graph Laplacian — as a market-stress indicator. Paper 3 uses *modularity* Q — a scalar derived from a partition of a thresholded correlation network. Modularity has an underlying matrix, the modularity matrix $\mathbf{B}$, whose eigenvalues and singular values nobody in this literature reports. Is there a study in comparing the Laplacian and the modularity matrix as sources of market indicators?

Section 5 argues that there is, that it is well-posed, and that the comparison is more interesting than it first looks because the two operators are *provably the same object on a regular graph* and yet behave completely differently on real market networks. Section 6 proposes three directions. Section 7 gives the preliminary numbers that motivate them, computed this week on data already in hand.

2 Timeline

Dates are the git commit dates in `Financial_networks_project`; every row corresponds to code and results in the repository, not to notes.

Date	Milestone	What happened
2026-07-05	Paper 1 complete	GMRF graph-Laplacian learning (Cardoso & Palomar). Solver, four experiments, figures, 769-line development log.
2026-07-22	Repo reorganised	Split into <code>paper1_gmrf_laplacian/</code> , <code>paper2_ticc/</code> , <code>paper3_market_modularity/</code> ; both source PDFs tracked.
2026-07-22	Paper 2 solver	TICC: Toeplitz graphical-lasso ADMM, dynamic-programming assignment, EM loop; synthetic generator, metrics, six baselines.
2026-07-22	Paper 2 experiments	Table 1, Table 2, Figs. 3–4, notebook.
2026-07-28	Paper 2 audit	Replication status reported honestly: three genuine non-reproductions identified and <i>not</i> absorbed into the headline.

Date	Milestone	What happened
2026-08-09	Paper 3 Phases 0–4	Universe and price panel; pre-registered crisis windows; returns and quality gates; 6,315 daily networks; Louvain communities, $k[t]$, $\Pi[t]$, three modularities; blockmodel and configuration nulls (126,300 generated graphs).
2026-08-10	Paper 3 Phases 5–9	Eight topological measures (132,615 graph evaluations); scoring with the overlap correction; PCA; extensions (crisis detector, event study, Louvain stability); replication report and notebook.
2026-08-10	Paper 3 re-audit	Adversarial re-audit of the replication itself: which non-reproductions are real, which are our artifacts. Eight defects found in our own criticisms.
2026-08-12	This document	Literature review on spectral stress indicators; problem statement; three directions; preliminary spectral probe on the Paper-3 network sequence.

3 Part I — What was built and what it showed

3.1 Paper 1: learning graph Laplacians from returns

Source. Cardoso & Palomar, *Learning Undirected Graphs in Financial Markets*, 54th Asilomar Conference on Signals, Systems and Computers, 2020, [arXiv:2005.09958](#). The extended treatment is [arXiv:2012.15410](#).

Method. A penalised maximum-likelihood estimate of a precision matrix constrained to be a valid graph Laplacian:

$$\min_{\mathbf{L}} \text{tr}(\mathbf{L}S) - \log \text{gdet}(\mathbf{L}) + h_{\alpha}(\mathbf{L}) \quad \text{s.t.} \quad \mathbf{L}\mathbf{1} = 0, \mathbf{L}_{ij} \leq 0 \ (i \neq j), \mathbf{L} \succeq 0, \quad (1)$$

where S is the sample correlation matrix and gdet is the pseudo-determinant (a valid Laplacian is singular, so $\det \mathbf{L} = 0$). The constraints have financial content: $\mathbf{L}\mathbf{1} = 0$ removes the market factor, and $\mathbf{L}_{ij} \leq 0$ forces all conditional correlations non-negative. Algorithm 1 adds a spectral penalty to force k connected components; Algorithm 2 rolls Algorithm 1 causally over 30-day windows with a temporal smoothness penalty $\delta \sum_t \|\mathbf{L}_t - \mathbf{L}_{t-1}\|_F^2$. The indicator is $\lambda_2(\mathbf{L}_t)$, the algebraic connectivity or Fiedler value [9], and the trading rule is: hold the portfolio when $\lambda_2 < \tau$, sit in cash otherwise, with $\tau = 1.0$.

Implementation. Rather than the paper’s ADMM, the solver works in edge-weight space: $\mathbf{L} = \mathcal{L}w$ with $w \geq 0$, which makes the Laplacian constraints automatic and turns the problem into a bound-constrained smooth minimisation solved by L-BFGS-B. Degree control ($\text{diag } \mathbf{L} = \mathbf{1}$) becomes a quadratic penalty; the k -component constraint becomes Fan’s spectral penalty with strength η .

Status.

- Fig. 1 (preprocessing) — **reproduced**. Panel (d), scaled data with the market factor *not* explicitly removed, is cleanest, as the paper claims.
- Fig. 2 (k -component vs. SGL) — **reproduced**. True $k = 3$ (rank 94 = $p - 3$ at $p = 97$) achieved, but only at $\eta = 300$; $\eta \in \{10, 150\}$ is insufficient. The paper does not report needing this.
- Fig. 3 (time-varying λ_2) — **reproduced**. Our $\lambda_2 \in [0.66, 1.50]$ matches the paper’s monthly regime structure.
- Fig. 4 (trading) — **partial**. *The mechanism replicates and the P&L ranking does not.* With the paper’s own $\tau = 1.0$ and evaluation window (ending 2020-05-01), the gate is live (79 of 201 days invested). S2 dodges the COVID crash exactly as the paper shows — flat at ≈ -0.01

while buy-and-hold falls to -0.21 — and also sits out autumn 2019. But our λ_2 stays near 1.3 through April where the paper’s drops below 1.0, so S2 misses the rebound: final S2 = -0.011 against S1 = $+0.078$. No threshold was tuned to fix this.

The most instructive failure. An early version of the temporal penalty used $S_{\text{aug}} = S_t + (2\delta/n_t)\mathbf{L}_{t-1}$. The Frobenius cross-term is $-2\delta \text{tr}(\mathbf{L}\mathbf{L}_{t-1})$, so the sign was inverted: the penalty *repelled* each window from its predecessor. The symptom was λ_2 collapsing into $[0, 0.78]$ with 65 windows at exactly zero (disconnected estimates), which made $\tau = 1.0$ non-binding and the trading figure vacuous. This is recorded because it is the kind of error that produces a plausible-looking figure: a stress indicator that spends its time near zero and spikes occasionally still *looks* like a stress indicator.

Worth keeping in mind for any new indicator build: check the proposed threshold actually gates before interpreting the backtest.

3.2 Paper 2: TICC

Source. Hallac, Vare, Boyd & Leskovec, *Toeplitz Inverse Covariance-Based Clustering of Multivariate Time Series Data*, KDD 2017, [arXiv:1706.03161](#) [10].

Method. Simultaneous segmentation and clustering: each cluster is a block-Toeplitz Gaussian MRF over a length- w window; the M-step is a Toeplitz graphical lasso solved by ADMM, the E-step an exact Viterbi assignment with a switching penalty β .

Status. The headline reproduces — TICC reaches macro-F1 ≈ 0.99 against the paper’s 0.95 average, and is genuinely robust across data seeds ($\sigma \leq 0.006$). Three things do not:

- The $\beta = 0$ **ablation is systematically too weak** (our 0.80 average vs. the paper’s 0.88), which means our replication *overstates* the value of the temporal-consistency penalty. This is the one place our error points away from the paper’s own conclusion, so it leads the report. **not reproduced**
- Network recovery (Table 2) sits at 0.80 against the paper’s 0.85 band. **partial**
- Fig. 4 is a timing proxy that never calls the fitting routine, over a narrower range of T than the paper’s 10^4 – 10^7 . **partial**

Known thumbs on the scale, disclosed: segment length 400 everywhere instead of the paper’s 100K (up to $2\times$ the data), and generator edge weights bounded away from zero. Window-size robustness, the micro-F1 cross-check and BIC selection of K are **not implemented**.

Relevance going forward. TICC is the one of the three that estimates a *regime*, not a snapshot. It is not central to the Laplacian-vs-modularity question, but it is the natural comparator if we ever want to ask whether a spectral indicator is doing anything a proper regime-switching model would not.

Do you want F per 2 carried forward at all, or parked?

3.3 Paper 3: modular dynamics of financial market networks

Source. Silva, Comin, Peron, Rodrigues, Ye, Wilson, Hancock & Costa, *Modular Dynamics of Financial Market Networks*, [arXiv:1501.05040](#) (2015) [29].

Method. Log returns; Pearson correlation over a $\Delta t = 30$ -return sliding window shifted by one day; a threshold $\tau(t)$ re-solved every day so that exactly $f = 10\%$ of pairs survive (fixed density, not fixed threshold); binarise. Louvain communities each day, giving $Q(t)$ and $k(t)$. Then the central construction: keep *only* the community sizes and the inter-community mixing matrix $\Pi(t)$, discard the network, and regenerate a random graph from that summary alone. The claim is that this compressed description reproduces the real network’s topology, and that crises move the market from well-defined communities to a uniform organisation.

Scale of the replication. $N = 360$ NYSE stocks $\times T = 6,345$ trading days (paper: $348 \times 6,008$), 6,315 daily networks, 126,300 generated null graphs, 132,615 graph evaluations across eight topological measures. Organised around a 26-item methodological audit written *before* implementation began, with crisis windows pre-registered in Phase 0 so that no detector could be tuned after seeing the data.

What reproduces. The community null tracks real modularity at $\rho = 0.98$ (paper: 0.99). Mean degree is exactly $f(N - 1) = 35.90$ every day. Crises shatter the network: Q falls to 0.0897 on 1987-10-19, the 25-year low, with $k \rightarrow 156$. The configuration-model noise floor is $Q = 0.1065 \pm 0.0145$, matching the ≈ 0.10 the paper draws but never states. Across all 32 cells of the paper’s correlation table the mean absolute difference is 0.155 and the sign agrees on 97%.

What the paper’s own construction does not support. These are the results worth carrying forward, and they are the reason the current problem statement exists.

1. **The level of Q is not evidence.** Running the identical pipeline on i.i.d. Gaussian returns of the same shape gives Louvain $Q = 0.2246 \pm 0.0067$ against the real market’s 0.2215 ± 0.0413 . Structureless data is, if anything, marginally *more* modular. Only the *variance* of Q distinguishes the market ($6\times$ that of noise). The mechanism is that a correlation matrix is positive semi-definite, so thresholding it yields a graph that is transitive by construction, and transitive graphs are modular; no degree-sequence null can control for this, because rewiring destroys the geometry along with the structure.
2. **The compression claim inverts exactly where the story lives.** The community summary costs $k + k(k + 1)/2$ numbers against the degree sequence’s N ; break-even is $k = 25.4$. It is cheaper on 69.3% of all days but only 34.3% of pre-registered sharp-crisis days, because the network shatters into singletons and k reaches 156.
3. **$\tau(t)$, which fixed-density thresholding throws away, detects crises far better than modularity does.** Against the pre-registered windows (2.7% base rate): mean correlation 0.807, $\tau(t)$ 0.807, isolated-node count 0.760, dynamical modularity **0.538**. The paper’s Fig. 1 has a box labelled “crisis detection evaluation” and the paper performs none; when it is built, the paper’s central quantity is the weakest of the four signals its own pipeline produces.
4. **Louvain instability rivals the signal.** Ten seeds per day: the same-day seed-to-seed spread in Q is $2.09\times$ the median day-to-day move and exceeds it on 77.9% of days. Same-day NMI between seeds is 0.543; consecutive-day NMI at fixed seed is 0.463. Two runs on identical data agree only +0.079 more than runs on different days.
5. **The eight measures are about three.** Mean $|\rho|$ between measures 0.607; three of eight principal components carry 90% of the variance.

The negative result, reported as prominently as the positive ones. The audit’s own central hypothesis was that every correlation in the paper is computed on 30-day windows slid by one day, so consecutive points share 29/30 of their data, and ≈ 200 effectively independent points are presented as 5,978. Recomputing on non-overlapping windows moves the mean ρ from 0.783 to **0.777**. The mechanism is real — a unit test shows two independent random walks correlating at $|\rho| > 0.3$ through the same windows — but overlap inflates correlations that are *spurious*, and these are not. Effective sample size falls from 6,315 to 211, so the confidence intervals should widen; the point estimates stand.

The re-audit (2026-08-10). The replication was then attacked in the same way it had attacked the paper. Results: all 32 paper cells were transcribed correctly, and the four non-reproduced cells survive nine implementation variants — they are real. But the *cause* is upstream of the null model: our networks fragment on 63% of days (99.3% post-2002, median 30 isolated stocks) where the paper implies fragmentation is a Black-Monday-only event. Degree-null modularity correlates -0.875 with the isolated-node count, and on fully connected days $\rho(Q_{\text{real}}, Q_{\text{conf}})$ falls from 0.579 to 0.187, near the paper’s +0.07. Eight defects were found in our own criticisms (matching-index definition, a circular rich-club normalisation, an over-stated NMSE claim, a noise-excess ratio computed against two different floors, among others). Unaffected and solid: the compression inversion, the Louvain instability, the overlap negative result, the noise-level control, and the crisis detector.

Deliberately out of scope, recorded so nothing is silently dropped: robustness to $\Delta t \in \{20, 60, 120\}$; market-mode removal before network construction; a mechanism for the paper’s own unexplained 2002 structural break. All three re-enter as first-class questions in Section 6.

4 Part II — What the three replications jointly establish

Stripped of the paper-by-paper detail, five things are now known well enough to build on:

- F1. Thresholding a correlation matrix manufactures the structure it then measures.** A PSD matrix thresholded at fixed density is transitive by construction; i.i.d. noise reproduces 80% of the market’s modularity excess over a degree-sequence null. This is not specific to Paper 3 — it applies to any paper in the thresholded-correlation-network literature, which is precisely the argument of the recent *Physics Reports* review [19].
- F2. Fixed-density thresholding discards the strongest crisis signal the pipeline produces.** $\tau(t)$ and mean correlation both reach AUC 0.807; they are available before any community is detected, at zero cost.
- F3. Scalar summaries of a partition are noisy and weak.** Louvain’s seed-to-seed jitter is twice the daily signal, and Q ’s crisis-detection AUC is 0.538 against a 2.7% base rate.
- F4. The Laplacian-based indicator of Paper 1 works through a specific and fragile mechanism.** λ_2 is only interpretable when the graph is connected and the estimation is regularised; both the sign bug and the disconnection episodes reduced it to noise.
- F5. The two literatures do not talk to each other.** Paper 1 estimates a Laplacian and reads its second-smallest eigenvalue. Paper 3 thresholds a correlation matrix and reads a scalar derived from a partition of it. Neither looks at the operator the other uses; neither reports a spectrum.

5 Part III — Problem statement: Laplacian versus modularity spectra

5.1 The two operators

Let $\mathbf{A}$ be the adjacency matrix of a network on N nodes with degrees d_i and m edges, $2m = \sum_i d_i$.

$$\text{Laplacian:} \quad \mathbf{L} = \mathbf{D} - \mathbf{A}, \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N, \quad (2)$$

$$\text{Modularity matrix:} \quad \mathbf{B} = \mathbf{A} - \frac{\mathbf{d}\mathbf{d}^\top}{2m}, \quad \mu_N \leq \dots \leq \mu_2 \leq \mu_1. \quad (3)$$

They are read from opposite ends. $\mathbf{L}$ is positive semi-definite; its small eigenvalues describe how close the graph is to falling apart, and λ_2 is the Fiedler value [9]. $\mathbf{B}$ is indefinite by construction ($\mathbf{B}\mathbf{1} = 0$, so it always has a zero eigenvalue with $\text{tr } \mathbf{B} < 0$); its *largest* eigenvalues describe community structure [23, 22], and — the part almost nobody uses — its *most negative* eigenvalues describe *anti*-community structure, groups that avoid each other [7].

5.2 The identity that makes the comparison non-trivial

For a connected d -regular graph, $\mathbf{L} = d\mathbf{I} - \mathbf{A}$ and $\mathbf{B} = \mathbf{A} - (d/N)\mathbf{J}$. The all-ones vector is the top eigenvector of $\mathbf{A}$ and is annihilated by both, so with ν_i the adjacency eigenvalues,

$$\lambda_2(\mathbf{L}) = d - \nu_2(\mathbf{A}), \quad \mu_1(\mathbf{B}) = \nu_2(\mathbf{A}) \implies \boxed{\lambda_2(\mathbf{L}) + \mu_1(\mathbf{B}) = d} \quad (4)$$

whenever $\nu_2 > 0$. On a regular graph the Fiedler value and the leading modularity eigenvalue are the same number, read backwards. A study that compared them there would be measuring nothing.

This is exactly why the comparison *is* interesting on market networks. Paper 3’s construction holds the *mean* degree at $f(N - 1) = 35.90$ on every single day, so the networks are as close to the regular case as any real construction gets — and yet, measured on all 6,315 days:

Quantity	Value
$\bar{d}$ (exact, every day)	35.900
Residual $\bar{d} - [\lambda_2(\mathbf{L}) + \mu_1(\mathbf{B})]$	9.49 ± 4.63
restricted to the 2,342 connected days	9.21 ± 3.60
on the largest connected component only	11.11 ± 5.31
$\text{corr}(\lambda_2(\mathbf{L}), \mu_1(\mathbf{B}))$	-0.187
connected days only	-0.119
largest connected component only	-0.169

The identity fails by 26% of the mean degree, and the two quantities are *near-orthogonal in practice*. That gap is not noise: it is degree heterogeneity and fragmentation, and it is measurable. **The research question is therefore not “which indicator is better” but “the two operators are duals in theory and independent in practice — what exactly is the difference measuring, and does it carry information about market stress?”** That is a well-posed question with a clean answer available.

5.3 Literature map

Laplacian-side indicators in finance. Cardoso & Palomar [2, 3] learn a Laplacian-constrained precision matrix and use λ_2 as the stress indicator; this is Paper 1. Kang, Yen & Cheong [11] build filtration networks from correlation distances and use the *maximum spectral gap* of the unnormalised Laplacian, $\max_{k \geq 2}(\lambda_{k+1} - \lambda_k)$, arguing that it is large during crashes and zero otherwise, and using it to assign crashes a *duration* rather than a date. A parallel strand uses aggregate spectral functionals rather than single eigenvalues: Laplacian energy [15] and the Laplacian-energy-like measure [16] as early-warning inputs. On the covariance side rather than the graph side, the absorption ratio of Kritzman, Li, Page & Rigobon [13] — the variance share of the leading principal components — is the best-established spectral systemic-risk measure in finance, and is a mandatory baseline for anything we propose.

Modularity-side theory. Newman introduced the modularity matrix and the leading-eigenvector method [23, 22]. Its spectral theory has since been worked out carefully: Bolla et al. [1] give conditions under which the modularity spectrum has at least one positive eigenvalue, i.e. a non-trivial community structure exists at all; Fasino & Tudisco [5, 6] analyse generalised modularity matrices and show that the number of positive eigenvalues upper-bounds the number of disjoint modules, and [7] use the extreme *negative* eigenvalues for anti-community detection. On the random-matrix side, Chauhan, Girvan & Ott [4] and Sarkar, Henderson & Robinson [28] show that the number of communities and the depth of a hierarchy are fingerprinted by eigenvalues separated from the bulk; Peixoto [24] computes the spectra of modular networks and identifies the point at which the isolated eigenvalues merge into the bulk; Nadakuditi & Newman [21] locate the resulting *detectability threshold* and show that spectral modularity maximisation is optimal in the sense that no method succeeds where it fails. For sparse graphs the non-backtracking operator [12] and the Bethe Hessian [26] push that threshold to the information-theoretic limit and are the correct comparators if we ever move to a sparse regime.

Correlation-specific null models. The single most relevant reference is MacMahon & Garlaschelli [17]: applying network community detection to correlation matrices is *intrinsically biased*, because the null model implicit in modularity (the configuration model) is inconsistent with the statistics of correlation matrices. They redefine the modularity matrix using a random-matrix-theory null, subtracting both the market mode and the Marchenko–Pastur bulk, so that the residual contains only genuine group structure. This is exactly the object Paper 3 needed and did not use — and it is the natural “correct” modularity matrix for our comparison. The classical RMT results it rests on are Laloux et al. [14] and Plerou et al. [25]. The recent *Physics Reports*

This framing is the part I would most like your action to. The alternative framing — a plain empirical horse race between indicators — is easier to execute but much less interesting and reviewers in this area have seen many horse races.

review by Masuda, Boyd, Garlaschelli & Mucha [19] surveys the whole “beyond-thresholding” agenda and is the best single framing citation for why Paper 3’s construction is the weak link.

Financial network structure, for context. The filtering tradition — minimum spanning trees [18] and planar maximally filtered graphs [30] — is the alternative to thresholding. Temporal community structure in markets was studied by Fenn et al. [8] using the multislice modularity of Mucha et al. [20]; Paper 3 cites that machinery and does not use it, which is the direct cause of its Louvain instability problem. Geometric alternatives (Ollivier–Ricci curvature) were proposed as fragility indicators by Sandhu, Georgiou & Tannenbaum [27] and are a good third baseline.

5.4 The gap

Putting the map together:

- The Laplacian spectrum has been used as a market indicator, in at least three different forms (λ_2 , the maximum spectral gap, Laplacian energy).
- The modularity *matrix* has a well-developed spectral theory, with results — structural eigenvalue counts, detectability thresholds, anti-community eigenvalues — that are exactly the sort of thing a market indicator would want.
- **Nobody has applied the modularity spectrum to markets.** Searches across the finance-network literature return modularity used only as a scalar Q from a partition. No paper reports $\mu_1(\mathbf{B})$, $\mu_N(\mathbf{B})$, the count of structural eigenvalues, or the spectral gap of $\mathbf{B}$ as a time series for a market.
- No paper compares the two operators on common data with a common scoring protocol, and the one theoretical fact that makes such a comparison sharp — Eq. (4) — appears not to have been used as an organising principle.

We are unusually well-placed to close this: the pipeline, the 6,315-day network sequence, the pre-registered crisis windows and the scoring code all already exist, and the audit already tells us where the traps are.

6 Part IV — Three proposed directions

They are ordered by how directly they answer your question. D1 is the core comparative study; D2 is the theoretical contribution that makes D1 mean something; D3 is the extension that uses the existing findings most aggressively. D1 and D2 naturally combine into a single paper; D3 could be a second.

6.1 D1 — The operator comparison: duality, and what breaks it

Question. $\lambda_2(\mathbf{L})$ and $\mu_1(\mathbf{B})$ are affine duals on a regular graph (Eq. 4) and near-orthogonal on real market networks ($r = -0.17$). Decompose the residual. Which component of it — degree heterogeneity, fragmentation, or the market mode — carries the market-stress information, and which operator exposes it more directly?

What we would build.

1. A single pipeline that, for each day, computes the full spectrum of a *family* of operators on the same network: $\mathbf{A}$, $\mathbf{L}$, the normalised Laplacian $\mathcal{L} = \mathbf{D}^{-1/2}\mathbf{L}\mathbf{D}^{-1/2}$, $\mathbf{B}$, the normalised modularity matrix, and — crucially — the RMT-corrected correlation modularity matrix of MacMahon & Garlaschelli [17], which needs no thresholding at all.
2. An exact decomposition of the duality residual $\bar{d} - [\lambda_2 + \mu_1]$ into a degree-heterogeneity term and a connectivity term, validated against the observed 9.49 ± 4.63 .

3. A common scoring harness — the Phase-0 pre-registered crisis windows, trailing- z detector, ROC-AUC, and out-of-sample chronological split — so that every operator is judged identically. This already exists and was used to produce the numbers in Section 7.
4. Robustness over the parameters Paper 3 never tested: $\Delta t \in \{20, 30, 60, 120\}$, density f , universe size, and with/without market-mode removal. This retires audit items A5 and A6 as a by-product.

Why it is publishable. It is not a horse race; it is a structural result with a horse race attached. The claim “these two indicators are the same statistic on a regular graph and independent on market networks, and here is exactly what the difference is” is checkable, surprising, and useful to the sizeable literature that picks one operator without argument.

Main risk. The decomposition may turn out to be dominated by fragmentation, which is an artifact of fixed-density thresholding rather than a property of markets. Mitigation: the largest-connected-component variant is already computed (residual 11.11 ± 5.31 , $r = -0.169$), so the finding survives removing fragmentation — but this needs to be established across constructions, not just this one.

6.2 D2 — Detectability: is there community structure to detect at all?

Question. Paper 3 estimates a 360×360 correlation matrix from 30 observations. The ratio $q = N/\Delta t = 12$ puts the Marchenko–Pastur band at $[6.07, 19.93]$, and the whole matrix has rank at most 29. *Under random-matrix theory, how many eigenvalues of the daily correlation matrix actually escape the noise bulk — and therefore how many communities is it even possible to detect?*

Preliminary answer, computed this week on all 6,315 days: excluding the market mode, the mean number of eigenvalues above the Marchenko–Pastur upper edge is **0.70**, with a maximum of 3. On a typical day there is *less than one* statistically distinguishable group mode. Louvain nonetheless returns a median of 9 communities every day, and the paper builds a 25-year narrative on their dynamics.

What we would build.

1. A *detectability phase diagram* over $(\Delta t, N, f)$: for each combination, where does the market sit relative to (a) the Marchenko–Pastur edge and (b) the Nadakuditi–Newman spectral detectability threshold [21] for the resulting network? This tells any practitioner in this literature whether their parameter choice is inside or outside the region where community structure is estimable.
2. A deterministic, algorithm-free replacement for $k(t)$: the count of structural eigenvalues of $\mathbf{B}$ ([1, 6]) and of RMT-escaping modes of $\mathbf{C}$, compared against Louvain’s $k(t)$ and its seed-to-seed spread. This directly retires the Louvain-instability problem (F3) rather than measuring it.
3. A statement of how much of Paper 3’s — and the wider literature’s — community dynamics survives once the estimable region is respected.

Why it is publishable. This is a methodological result the thresholded-correlation-network literature needs and does not have; it is the constructive version of the “beyond thresholding” argument [19]. It also converts our strongest replication finding — that i.i.d. noise reproduces 80% of the market’s modularity excess — from a simulation into a theorem-backed statement.

Main risk. The answer may be uninteresting in one direction: if essentially *no* configuration used in the literature is inside the detectable region, the contribution is purely negative. That is still worth publishing, but it is a different (and harder to place) paper. Mitigation: the phase diagram will identify *which* $(\Delta t, N)$ combinations are usable, which is constructive.

6.3 D3 — The negative spectrum: anti-communities and the 2002 break

Question. The Laplacian is blind to sign and the thresholded network deletes every negative correlation. Paper 3’s construction discards 4,246,798 pair-days with $\rho < -\tau$ — strong anticorrelations recorded identically to no relationship at all — and that series *collapses to near zero from roughly 2003 onward and never recovers*, close to the paper’s own unexplained 2002 structural break, which it lists as future work. The modularity matrix, unlike the Laplacian, has a meaningful negative spectrum: $\mu_N(\mathbf{B})$ and its eigenvector identify anti-communities [7]. *Does the negative spectrum recover the discarded signal, and does it explain the 2002 break?*

Preliminary: $\mu_N(\mathbf{B})$ detects the pre-registered crises at ROC-AUC **0.760** — better than every Laplacian statistic we computed, and far better than Q ’s 0.538.

What we would build. Signed correlation networks with no thresholding step; the signed Laplacian and SPONGE-style generalised eigenproblems as comparators; the negative modularity spectrum as the indicator; and a formal change-point analysis of the 2002–2003 transition in the sign structure, against candidate mechanisms (decimalisation in 2001, ETF and index-fund growth, algorithmic trading).

Why it is publishable. It answers a question the original authors explicitly left open, using a quantity they structurally could not see.

Main risk. $\mu_N(\mathbf{B})$ correlates 0.68 with the market-mode signal, so part of its 0.760 is not new information (see Section 7). The direction is only worth pursuing if the *residual* after projecting out the market mode still carries signal — which is a two-day check, and should be the gate before committing.

6.4 Recommendation

D1 + D2 as one project, D3 gated on a quick check. D1 alone risks being a horse race; D2 alone risks being a negative result; together they are a single coherent claim: *here is the correct operator to read market community structure from, here is the region of parameter space where reading it is meaningful at all, and here is what the field’s two incumbent indicators are actually measuring*. D3 is a good second paper and a bad first one.

7 Part V — Preliminary evidence

Computed 12 August 2026 on the existing Paper-3 network sequence: $N = 360$ stocks, 6,315 daily networks, 1987–2011. For each day, the full spectrum of $\mathbf{L}$, of $\mathbf{B}$, and of the raw correlation matrix $\mathbf{C}$ was computed (three 360×360 symmetric eigendecompositions per day). Every signal was then scored through the *identical* detector already used in the replication: trailing 250-day z -score using only data strictly before the day scored, ROC-AUC against the Phase-0 pre-registered *sharp*-tier crisis windows (base rate 2.68%). The sign of each detector (spike vs. drop) was chosen to favour it, which is generous and is noted as such.

Signal	Operator	AUC (sharp)
Spectral entropy of $\mathbf{C}$ (effective rank)	correlation	0.822
$\lambda_1(\mathbf{C})/N$ (absorption ratio)	correlation	0.812
$\tau(t)$ (fixed-density threshold)	—	0.807
$\mu_N(\mathbf{B})$ (anti-community edge)	modularity	0.760
Isolated-node count	Laplacian null space	0.756
$\sum_{\mu_i > 0} \mu_i$ (positive modularity mass)	modularity	0.750
$\lambda_N(\mathbf{L})$ (largest Laplacian eigenvalue)	Laplacian	0.731
$\#\{\mu_i > 0\}$ (structural eigenvalue count)	modularity	0.722
$\lambda_2(\mathbf{L})$ on the largest component	Laplacian	0.652
$\mu_1 - \mu_2$ (modularity spectral gap)	modularity	0.636
$\lambda_2(\mathbf{L})$ (Fiedler value, whole graph)	Laplacian	0.617
$\mu_1(\mathbf{B})$ (leading modularity eigenvalue)	modularity	0.607
$\#$ modes above Marchenko–Pastur edge	correlation	0.619
$Q(t)$ (Louvain modularity — the paper’s signal)	partition scalar	0.538
$\max_k(\lambda_{k+1} - \lambda_k)$ (max Laplacian gap)	Laplacian	0.526

Readings.

1. **The modularity matrix beats the modularity scalar decisively.** $Q(t)$ scores 0.538; three separate statistics of the matrix it is derived from score 0.72–0.76. The information is in the operator; collapsing it to one number through a stochastic partition destroys most of it. *This is the direct empirical answer to your question, and it is a clean one.*
2. **The Laplacian statistics are weak on this construction.** The Fiedler value reaches 0.617, and 0.652 once fragmentation is removed by restricting to the largest component. Note that this is *not* a refutation of Paper 1: it uses a learned, regularised Laplacian on 7 stocks, not a thresholded 360-node network. The maximum-spectral-gap indicator of Kang et al. [11] scores 0.526 — essentially chance — but again on a different construction from theirs (fixed density, not distance filtration), so this is a caution to check, not a contradiction.
3. **Everything above 0.75 is one signal.** $\tau(t)$, the absorption ratio and the spectral entropy of $\mathbf{C}$ are pairwise correlated at $|r| > 0.99$; the isolated-node count correlates 0.92 with τ . They are all measuring market-mode dominance. This matters: the “best” indicators in the table are not four discoveries, they are one, and it is the oldest one in the finance literature [13, 14].
4. **The modularity family is a genuinely different axis.** $\mu_1(\mathbf{B})$ correlates 0.025 with the market-mode detector — essentially orthogonal — and 0.761 with Louvain Q in levels. So $\mu_1(\mathbf{B})$ is a deterministic proxy for what Louvain is trying to measure, without Louvain’s 2×-the-signal seed jitter.
5. **But orthogonality did not translate into out-of-sample gain.** Fitting a logistic detector on a pre-2002 training period and testing post-2002 (27 sharp-crisis days in the test set): market mode alone 0.730; adding $\mu_1(\mathbf{B})$ 0.629; adding Q 0.699; adding $\mu_N(\mathbf{B})$ 0.719. In-sample all of these *improve* on the market mode (0.812 $\rightarrow$ 0.822–0.831), which is exactly the pattern you would expect from overfitting. **With 27 out-of-sample events, none of this is conclusive in either direction** — it is a warning that the incremental-value claim needs a much larger event set (more markets, more assets, longer history) before it can be made.

The detectability number. Excluding the market mode, the mean count of correlation eigenvalues above the Marchenko–Pastur edge is 0.70 (max 3, often 0) at $\Delta t = 30$, $N = 360$. This is the single most consequential preliminary number in this document: it says that at the window length used throughout this literature, the daily correlation matrix contains *essentially no statistically resolvable group structure*, while the methods applied to it confidently report nine communities a day.

Reproducibility. Everything in this section is in `reports/probe/`: three scripts (`spectral_probe.py`, `duality.py`, `incremental.py`), the per-day spectra (`spectral_probe.parquet`, 6,315 rows $\times$ 15 columns) and the AUC tables as JSON. They import the replication’s own `crisis_dates`, `construct` and scoring code unchanged — the same pre-registered windows, the same 250-day trailing z -score — so every number above is directly comparable to the replication’s own. As a check on that claim, the probe’s $Q(t)$ and $\tau(t)$ AUCs reproduce the replication’s stored Phase-8 values (0.5380684370 and 0.8071316467) to machine precision, so the new spectral rows sit on exactly the same scoring protocol as the published ones. Runtime is about 6 minutes for the full spectral pass on one core.

8 Part VI — Decisions I would like your view on

- Q1. Framing.** Is the duality framing (Eq. 4) the right spine for the study, or would you prefer the plainer empirical comparison?
- Q2. Scope of the network construction.** Do we keep Paper 3’s fixed-density thresholded networks as the substrate — which makes everything directly comparable to the replication — or move to the threshold-free RMT-corrected correlation modularity of MacMahon & Garlaschelli [17], which is more defensible but breaks comparability? My inclination is to do both and report the difference, since that difference is itself the “beyond thresholding” argument made concrete.
- Q3. Data.** The current universe is 360 NYSE survivors, 1987–2011, and carries a double survivorship filter (documented). For the incremental-value question we need more crisis events than 27 out-of-sample days. Options: extend to 2026; add other exchanges; use index-level rather than stock-level data. Which do you consider most defensible?
- Q4. Target venue.** The framing changes with the venue — a physics venue (*Phys. Rev. E, Physica A*) wants the spectral/RMT result; a finance venue wants the indicator to beat the absorption ratio out of sample, which on current evidence it does not.
- Q5. Paper 2.** Carry TICC forward as a regime-model comparator, or park it?

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Appendix A — Environment and reproduction

- Repository: `Financial_networks_project`, branches `main`, `add-ticc-replication`, `add-silva-replication`.
- Interpreter for Paper 3: `/opt/anaconda3/bin/python` (3.12.4). `python-igraph` is a hard dependency (Louvain, exact clique number, betweenness, degree-preserving rewiring); the project `.venv` lacks `yfinance` and `pyarrow`.
- Every stochastic operation takes an explicit seed and logs it (`DATA_SEED = 7`, `FIT_SEED = 42`); null-model runs are bit-reproducible at any worker count.
- Full accounting lives in `paper1_gmrf_laplacian/dev_log.md` (769 lines), `paper2_ticc/CLAUDE.md`, `paper3_market_modularity/PITFALLS.md` (26-item audit) and `paper3_market_modularity/results/REPLICATION_REPORT.md`.

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